

Living IoT Simulator

Introduction

This simulator implements a bio-inspired wireless localization system based on the "Living IoT" concept presented in the SIGCOMM'19 paper: [Living IoT: A Flying Wireless Platform on Live Insects](https://arxiv.org/pdf/1812.09419) (<https://arxiv.org/pdf/1812.09419>).

The system utilizes live insects (bumblebees) as mobile platforms carrying IoT devices, with multiple Access Points (APs) employing beamforming techniques for real-time positioning.

Key features of the simulator:

- **Insect Behavioral Modeling:** Simulates bee movement patterns including foraging behavior, nectar collection, and hive return logic
- **Beamforming Localization:** Implements TDMA-based phase sweeping with configurable parameters
- **Multi-AP Coordination:** Supports 2-antenna APs with directional beam patterns and phase difference calculations. In future, simulations with more antennas will be implemented.
- **Signal Propagation:** Models preamble detection and signal attenuation due to obstacles. Multi-path effects will be implemented in future.
- **Data Visualization:** Generates real-time trajectory plots and angle estimation comparisons.

The system demonstrates how millimeter-scale backscatter devices mounted on insects can achieve:

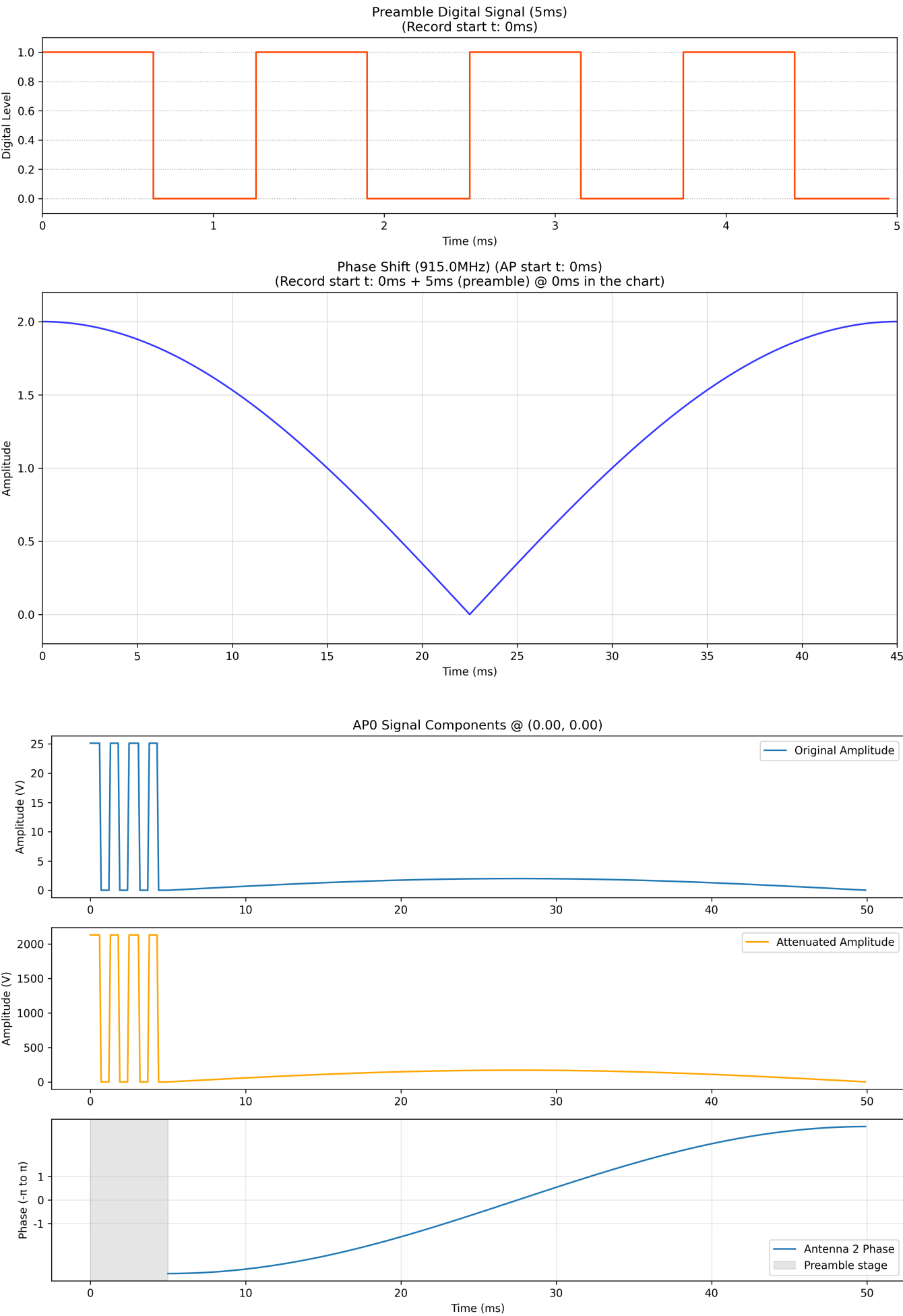
- Energy-efficient communication via beamformed signals
- Environmentally adaptive movement patterns
- Multi-AP collaborative positioning

Typical use cases include precision agriculture monitoring, search-and-rescue operations, and ecological research where GPS is unavailable or impractical.

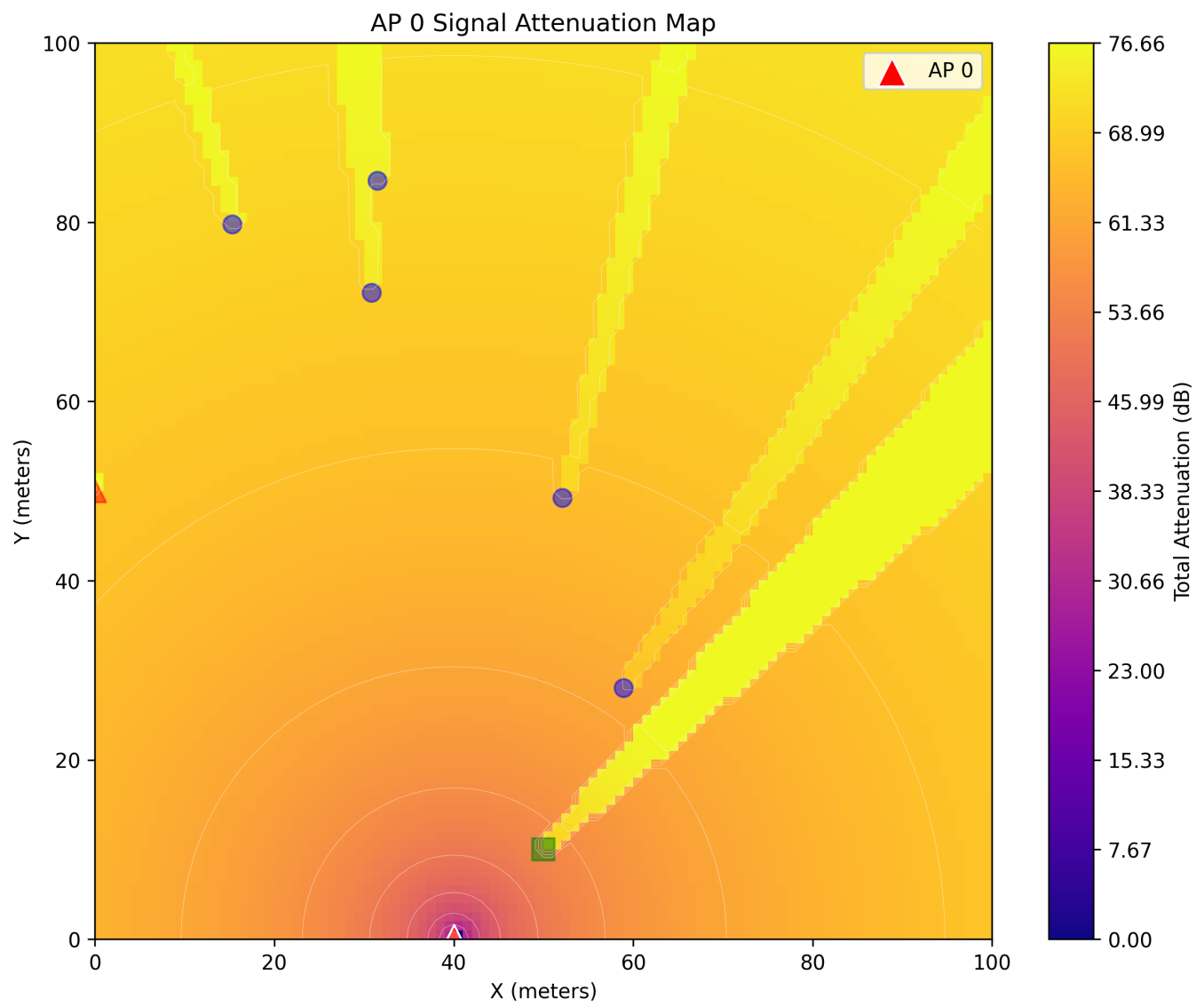
Features

Sample figures generated by the simulator:

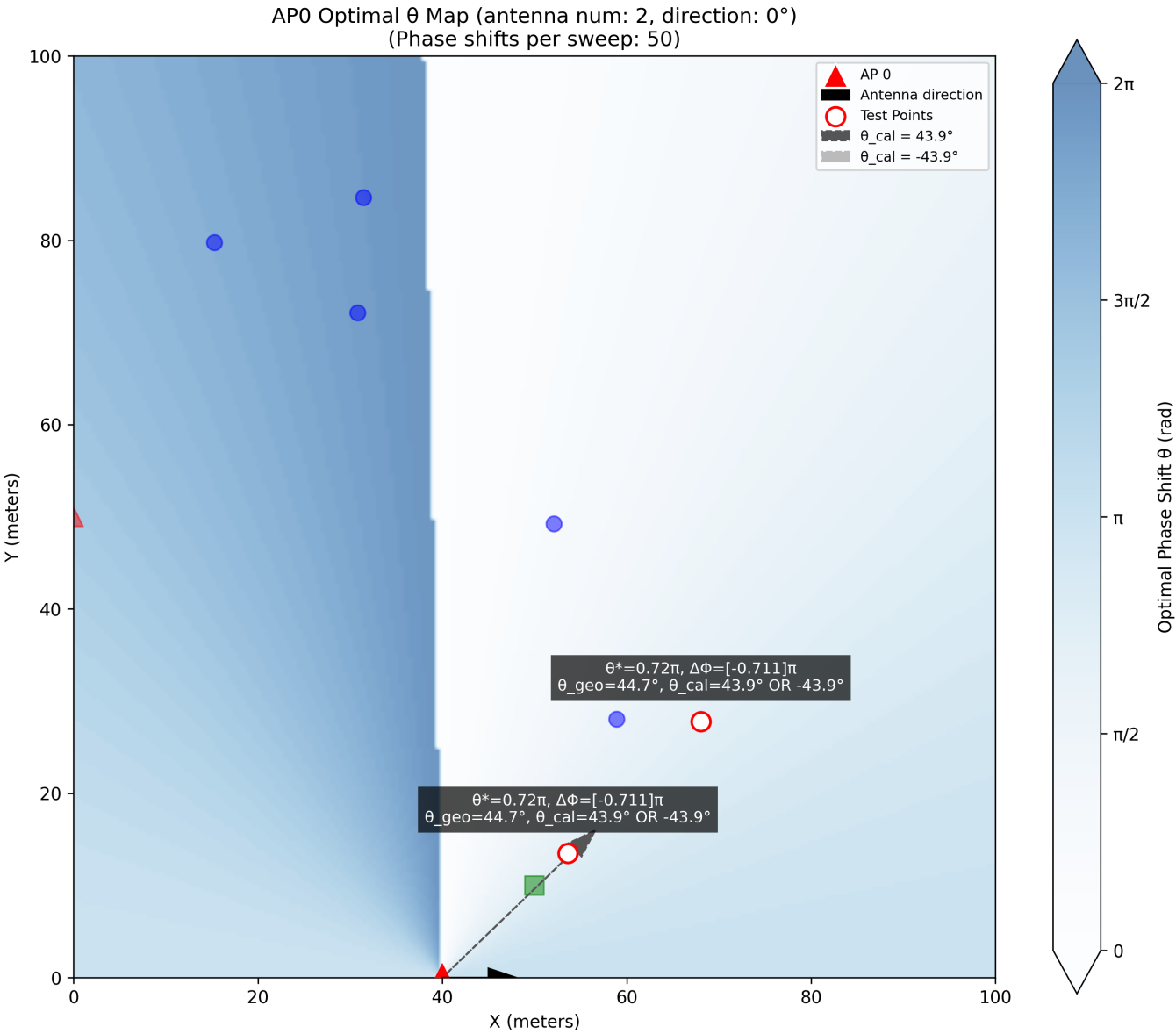
- Access Point (AP) signal characteristics:



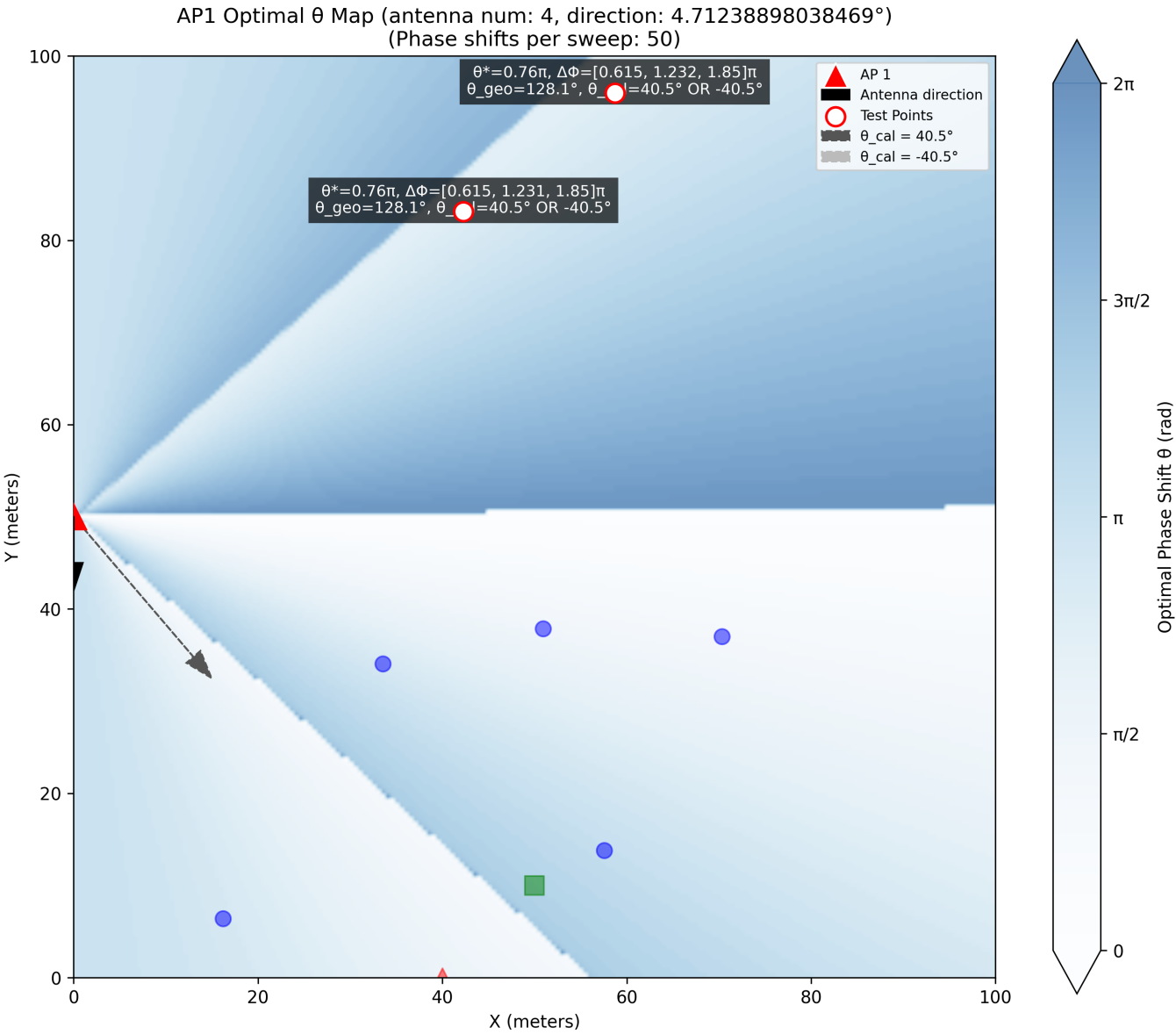
- Access Point (AP) signal static attenuation map:



- Access Point (AP) antenna beamforming best theta estimation pattern:

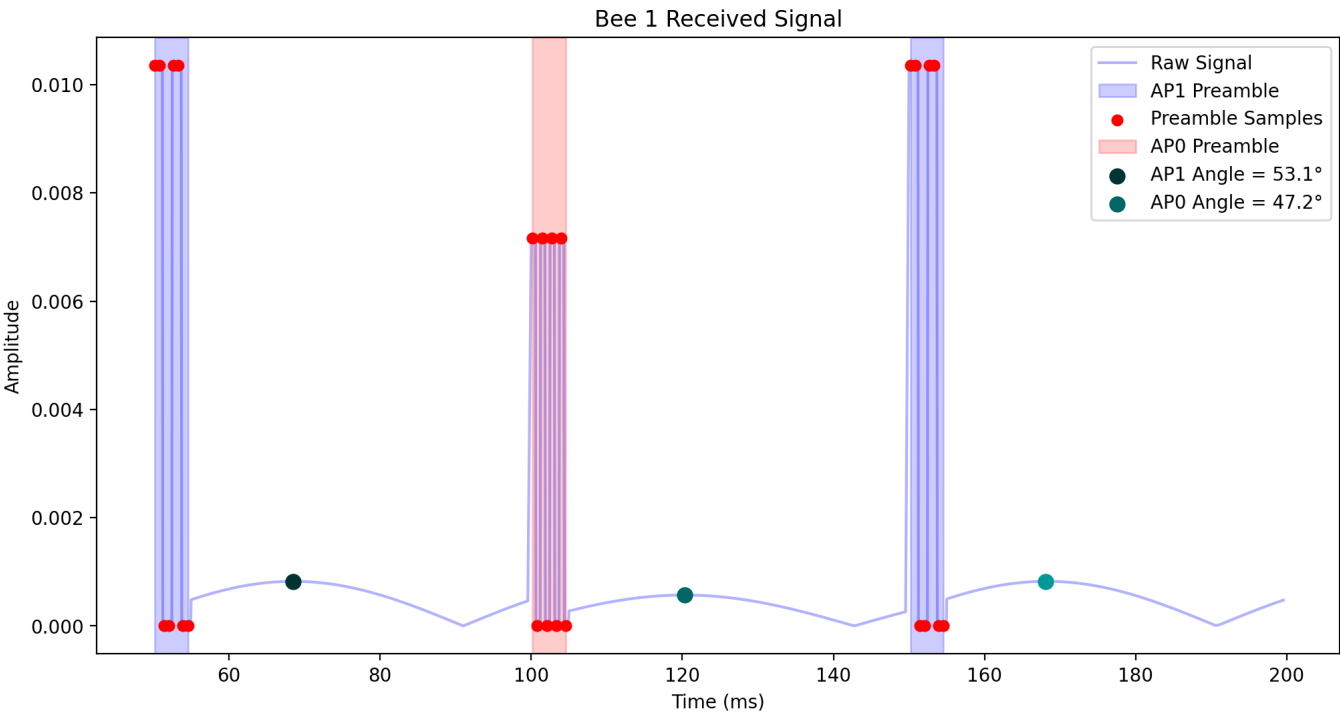
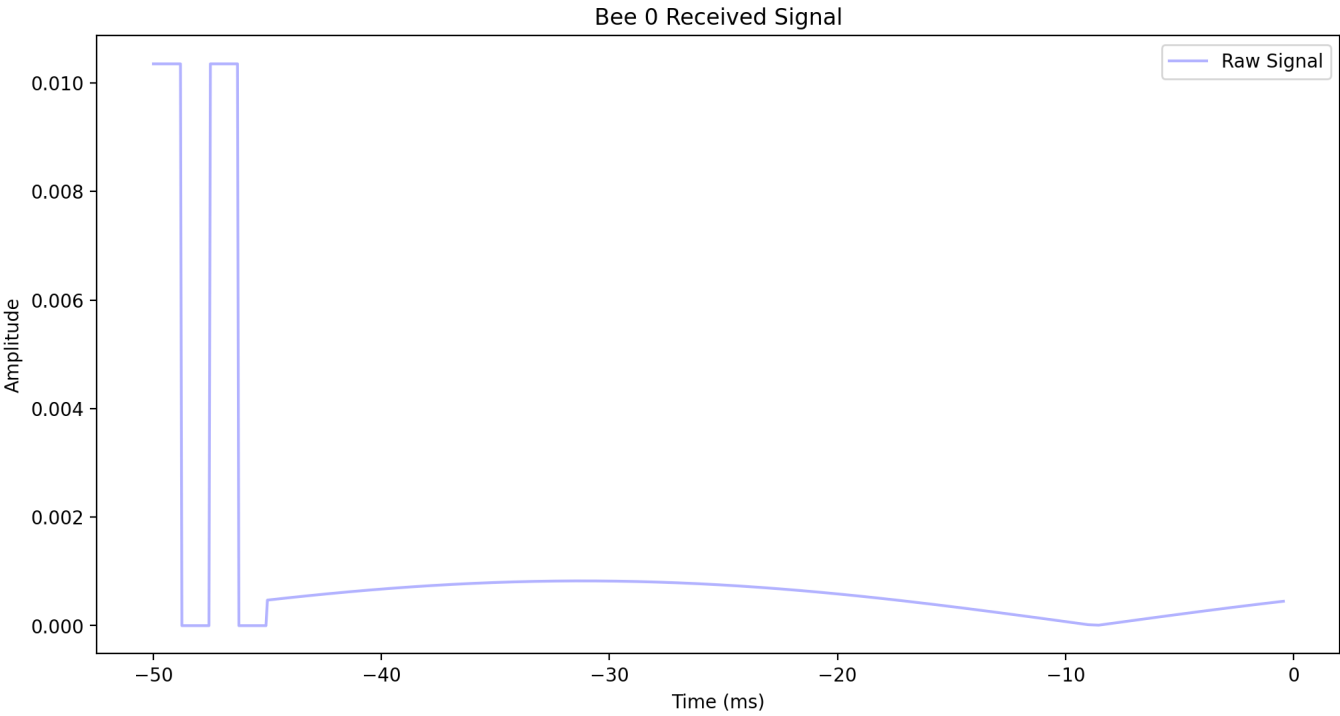


Note that when there are more than 2 antennas deployed, there may be more than one possible direction with the same optimal theta sampled. For example, when there are 4 antennas deployed, the IoT platform failed to detect the AP orientation (thinking in AP's perspective):

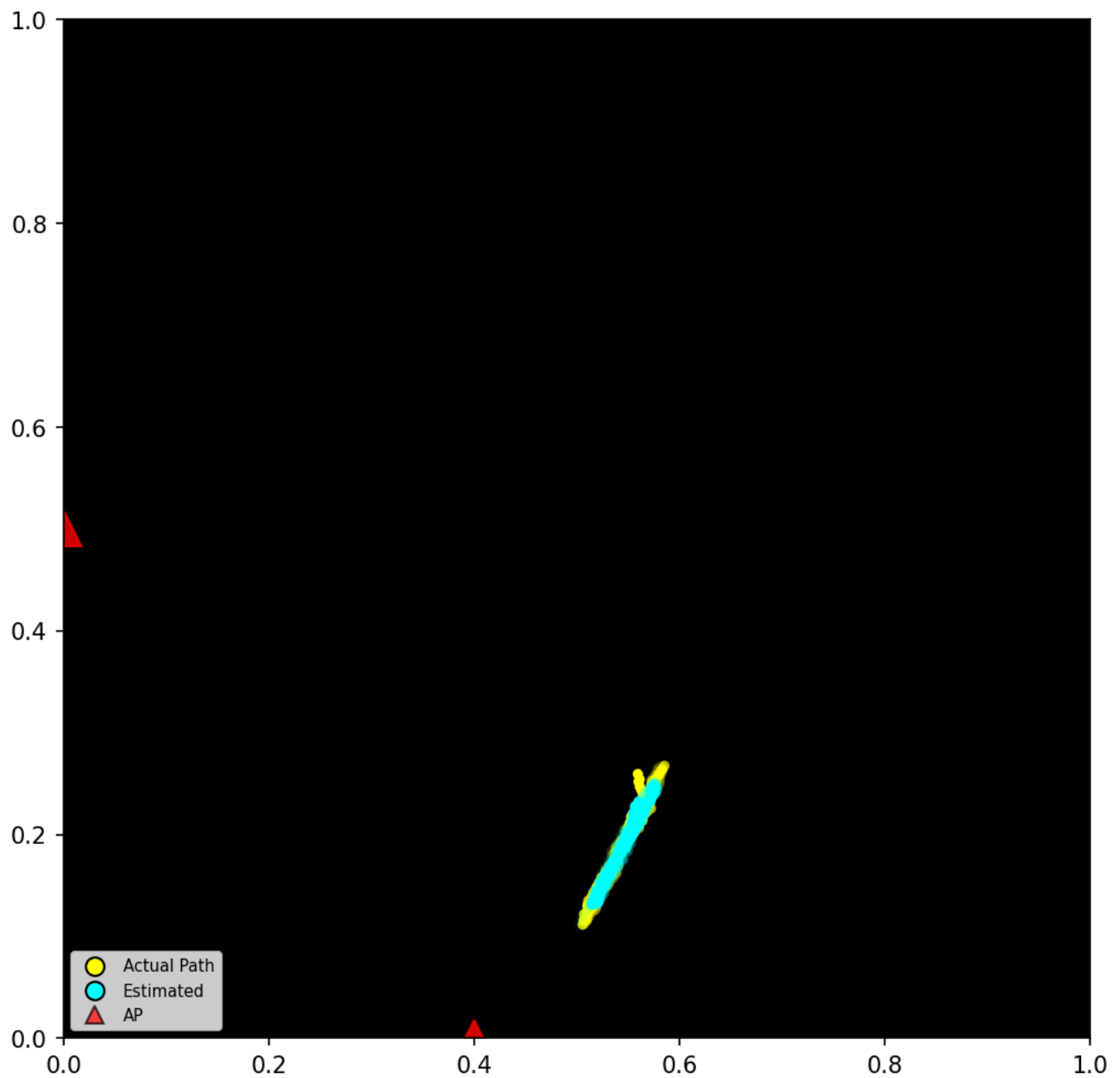


More efficient co-judgement algorithms will be implemented in the near future.

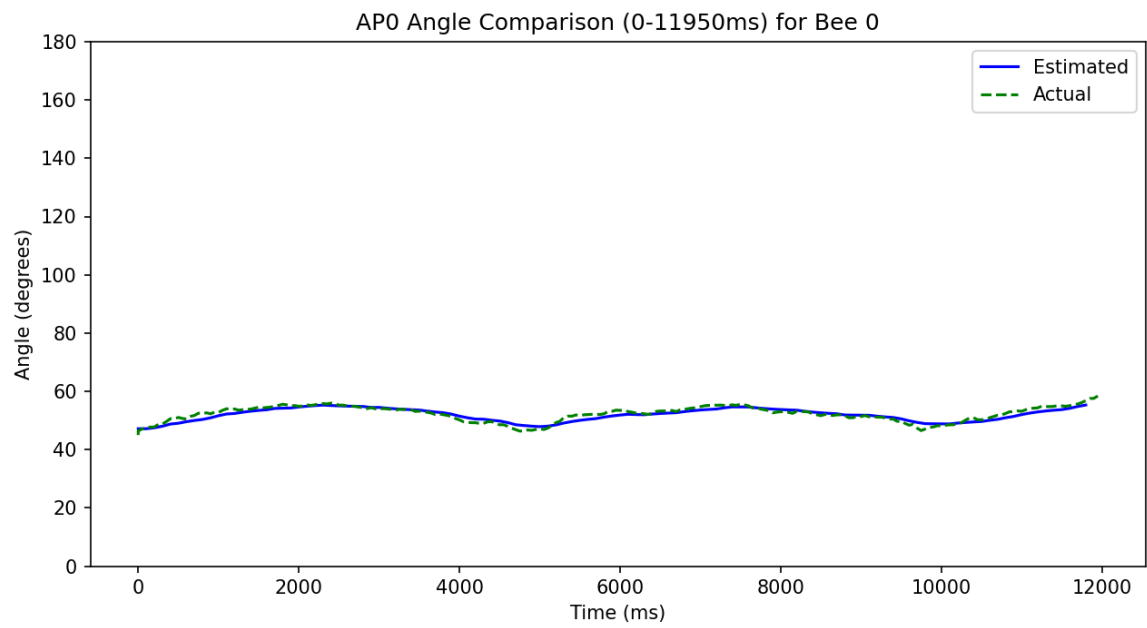
- Bee received raw signals and processed signals from APs:



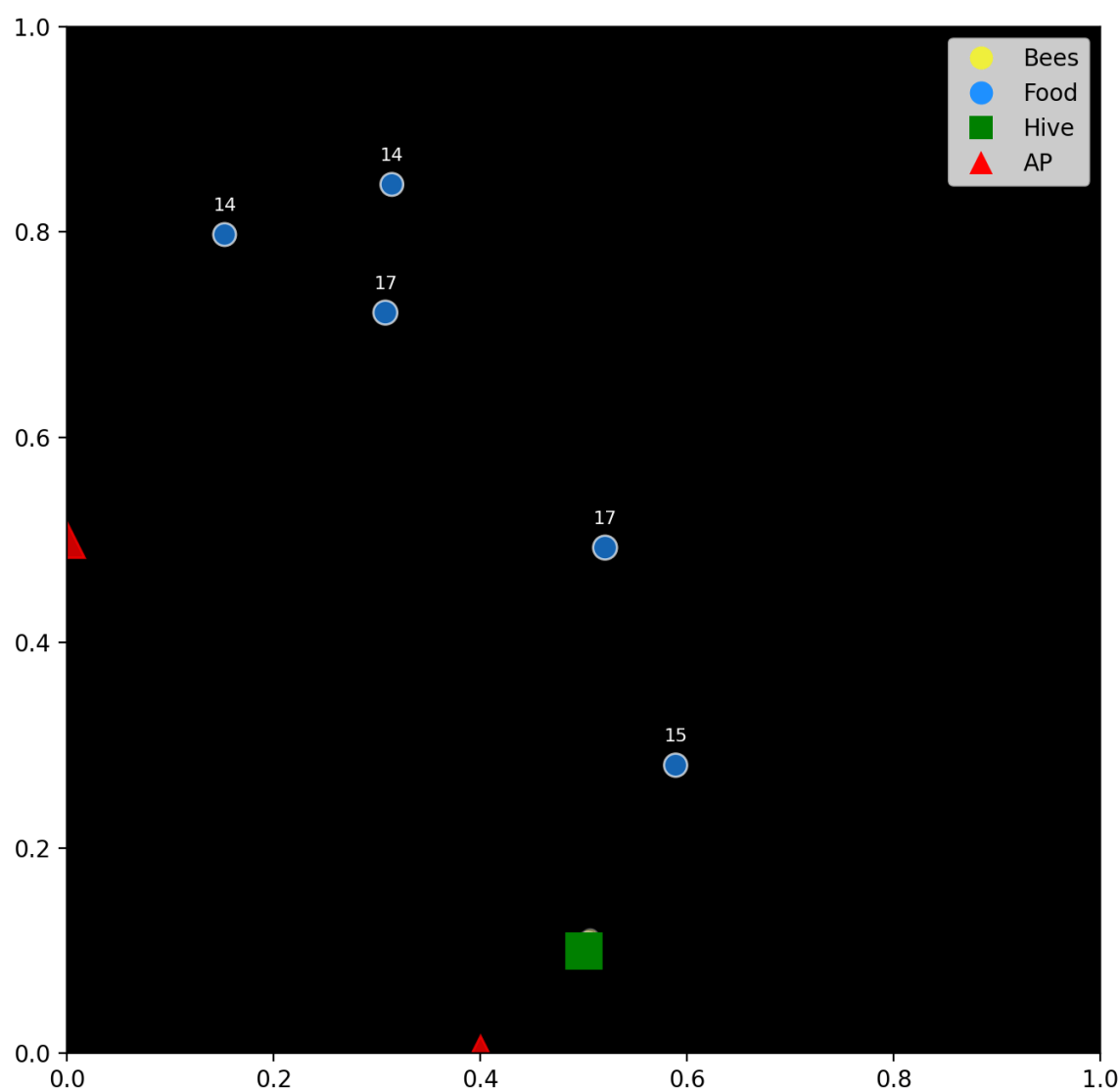
- Bee trajectory and estimated trajectory when there are 2 antennas deployed:



- Bee angle estimation comparison between the proposed method and the ground truth in AP's perspective:



- The overall virtual scene throughout the simulation:



Besides the figures, several important simulation statistics are also generated with examples in .csv files, including:

- Static, discretized AP NLOS calculation results
- Bee trails and estimated trails in AP's perspective
- The received raw signals from APs in Bee's perspective

Quick Start

- Install the required packages:

```
pip install -r requirements.txt
```

- Run the simulator:

```
python simulator.py
```

Future work

- Add more simulation analysis and plots.
- Address the case where more than 2 antennas are deployed.
- Add more noise effects and real-scenario modelling tests.
- Implement Chapter 4.1: Self-Localization Accuracy
- Implement Chapter 3.2.2: Addressing Multi-path Effect
- Implement Chapter 3.3: Backscatter Communication Design

Credits

- The simulator is developed by Kunhuan Wu (24137022g@connect.polyu.hk). Welcome to contact me for any questions and feedback!